Biomedical Signal Acquisition using NI-ELVIS II^{+} and i-Worx

1. Introduction to NI ELVIS II+

NI ELVIS II+ (National Instruments Educational Laboratory Virtual Instrumentation Suite) is a combined hardware and software platform used for hands-on engineering and biomedical instrumentation education. It functions as a data acquisition (DAQ) device, connecting physical sensors to a computer so that signals can be recorded, processed, and analyzed in real time.

1.1 Acquisition Software: LabScribe vs. MATLAB

iWorx sensors are normally paired with LabScribe, the manufacturer's own dedicated data-acquisition and analysis software. LabScribe provides a ready-made interface for recording, viewing, and doing basic analysis on iWorx sensor signals without requiring any custom programming.

In this project, however, all acquisition, filtering, and analysis was instead implemented directly in MATLAB rather than LabScribe. This approach was chosen to allow full custom control over the signal processing pipeline (specific filter designs, real-time peak/arrhythmia detection, custom calibration routines, and fully custom GUIs) which is not possible within LabScribe's fixed interface. This required bypassing LabScribe entirely and communicating with the NI ELVIS hardware directly at the driver level from MATLAB.

1.2 Integrating MATLAB with NI ELVIS II+

To allow MATLAB to talk directly to the NI ELVIS II+ hardware (instead of going through LabScribe), the following software components were installed and used:

- **NI-DAQmx driver:** the low-level National Instruments driver that allows any software (including MATLAB) to detect and communicate with NI DAQ hardware such as the ELVIS II+. Without this installed, MATLAB cannot see the device at all.
- **MATLAB Data Acquisition Toolbox:** provides the daq.createSession('ni') based session API used throughout this project (the "legacy" session interface, required for compatibility with MATLAB R2018b).
- **NI-DAQmx Support Package for MATLAB**: an add-on installed via MATLAB's Add-On Explorer that links the Data Acquisition Toolbox to the NI-DAQmx driver, exposing the ELVIS II+ analog input channels (ai0, ai1, ai2, ...) to MATLAB as addressable channels.
- **MATLAB Signal Processing Toolbox:** supplies the filter design and analysis functions used across all experiments.

Once installed, the device is verified and enumerated in MATLAB using daq.getDevices, which confirmed the ELVIS II+ was recognized as device 'Dev1' on the system used for these experiments. From that point, MATLAB could create a session, add analog input channels tied to specific iWorx sensor outputs, and stream data directly into custom-built real-time GUIs, completely replacing LabScribe's role.

Detecting the DAQ Device & Identifying Analog Channels in Windows (NI MAX)

Before writing any MATLAB code, the NI ELVIS II+ was first detected and verified in Windows using NI Measurement & Automation Explorer (NI MAX), the diagnostic utility installed alongside the NI-DAQmx driver. Once connected via USB, the device appears in NI MAX under "Devices and Interfaces" as "NI ELVIS II+: Dev1", confirming Windows has recognized the hardware before MATLAB ever tries to access it.

NI MAX also provides a Test Panels feature, used here to identify and verify which physical analog input channel (e.g. ai2) a given sensor was actually wired to, before trusting that assignment in code. Opening Test Panels -> Analog Input, selecting the channel under test, and setting Input Configuration to Differential shows a live "Amplitude vs. Samples" chart. With the FT-220 sensor connected, channel ai2 showed a clear, live pulsatile waveform, confirming the sensor was correctly wired to that channel before any MATLAB script was written.

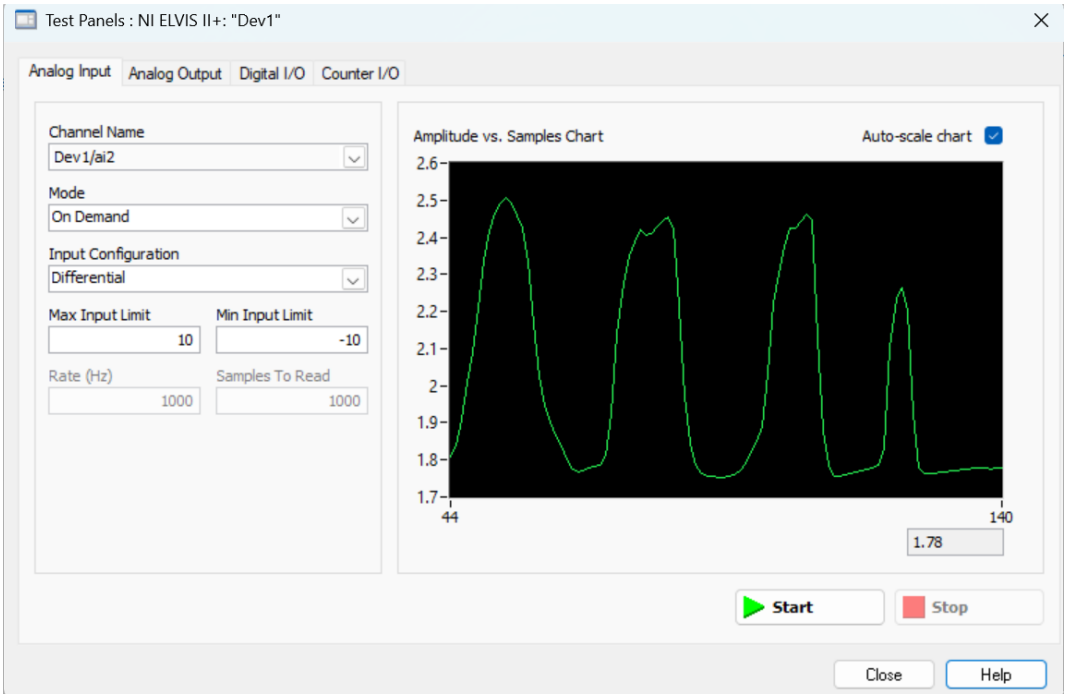


Figure: NI MAX Test Panels — Analog Input test on Dev1/ai2, Differential mode, showing a live waveform confirming correct channel wiring.

Adjusting Hardware Filter/Gain Values on the IX-ELVIS Board

The iWorx system and NI ELVIS II+ acquire physiological signals using manufacturer-designed analog front-end circuitry. The analog filtering stages (such as anti-aliasing and conditioning filters) are implemented in hardware with fixed characteristics determined by the manufacturer and are not user-adjustable during data acquisition. Consequently, the signal received by the analog-to-digital converter (ADC) has already undergone the necessary hardware-based conditioning and generic filtering required for reliable digitization. After the signal is digitized, additional digital signal processing can be performed in MATLAB, where custom digital filters (e.g., Butterworth band-pass, low-pass, or notch filters) may be applied to further remove noise or isolate the frequency components of interest. By using LABVIEW, the amplification (gain) of the acquired signal can be adjusted through the acquisition interface by configuring the amplifier settings, allowing optimization of the signal amplitude also alter the built-in analog filter characteristics.

1.3 NI ELVIS II+ Base Unit

The NI ELVIS II+ base unit (the white housing) is a multi-instrument platform that combines several pieces of standard lab equipment into one device, controlled through USB from a PC. Onboard instruments relevant to this project include:

- Oscilloscope (Scope) inputs: CH0 and CH1, used for direct analog signal viewing.
- Function Generator (FGEN): a programmable/adjustable waveform generator, with dedicated front-panel knobs for frequency and amplitude control.
- Variable Power Supplies: adjustable DC voltage supplies used to power auxiliary circuitry on the prototyping board.
- Digital Multimeter (DMM) inputs: VΩ, COM, and A terminals, for voltage, resistance, and current measurement.
- A prototyping breadboard area, allowing custom circuits to be physically wired alongside the fixed instrumentation.

In this project, the ELVIS II+ base unit's analog input channels served as the physical entry point for each iWorx sensor's output voltage, which was then digitized and streamed into MATLAB.

1.4 IX-ELVIS Interface Board (iWorx)

Mounted directly on top of the NI ELVIS II+ prototyping area is the IX-ELVIS interface board (manufactured by iWorx Systems), visible as the blue PCB in Figure 1.1. This board acts as the bridge between raw iWorx sensor leads and the ELVIS II+ analog inputs, and contains the analog front-end signal conditioning circuitry for each channel:

- Instrumentation Amplifiers (InAmp): first-stage amplifiers for each channel, boosting the raw, low-amplitude sensor signal (particularly critical for ECG, which is only millivolts in amplitude at the source).
- Programmable Gain Amplifiers (PGA): allow the overall channel gain to be adjusted to suit different sensor types and signal strengths.
- High-Pass Filters (HPF) and Low-Pass Filters (LPF): fixed analog filter stages per channel that perform basic hardware-level noise/drift rejection before the signal ever reaches MATLAB's software filters.
- Dual Buffer stages: signal buffering circuitry for channels 5-8.
- Onboard power regulation: supplies regulated +7.0 VDC, -7.5 VDC, and -5.6 VDC rails (plus ground) to power the analog circuitry on the board.
- A prototyping breadboard section, allowing additional custom circuitry to be added alongside the fixed sensor-conditioning stages.

Each iWorx sensor lead (ECG electrodes, PTN-104, SPN-304, or the dynamometer) connects to this board via a dedicated BIO channel connector, passes through its InAmp -> PGA -> HPF/LPF chain, and the resulting conditioned analog voltage is what MATLAB ultimately samples through the ELVIS II+'s analog input.

1.5 Labeled Hardware

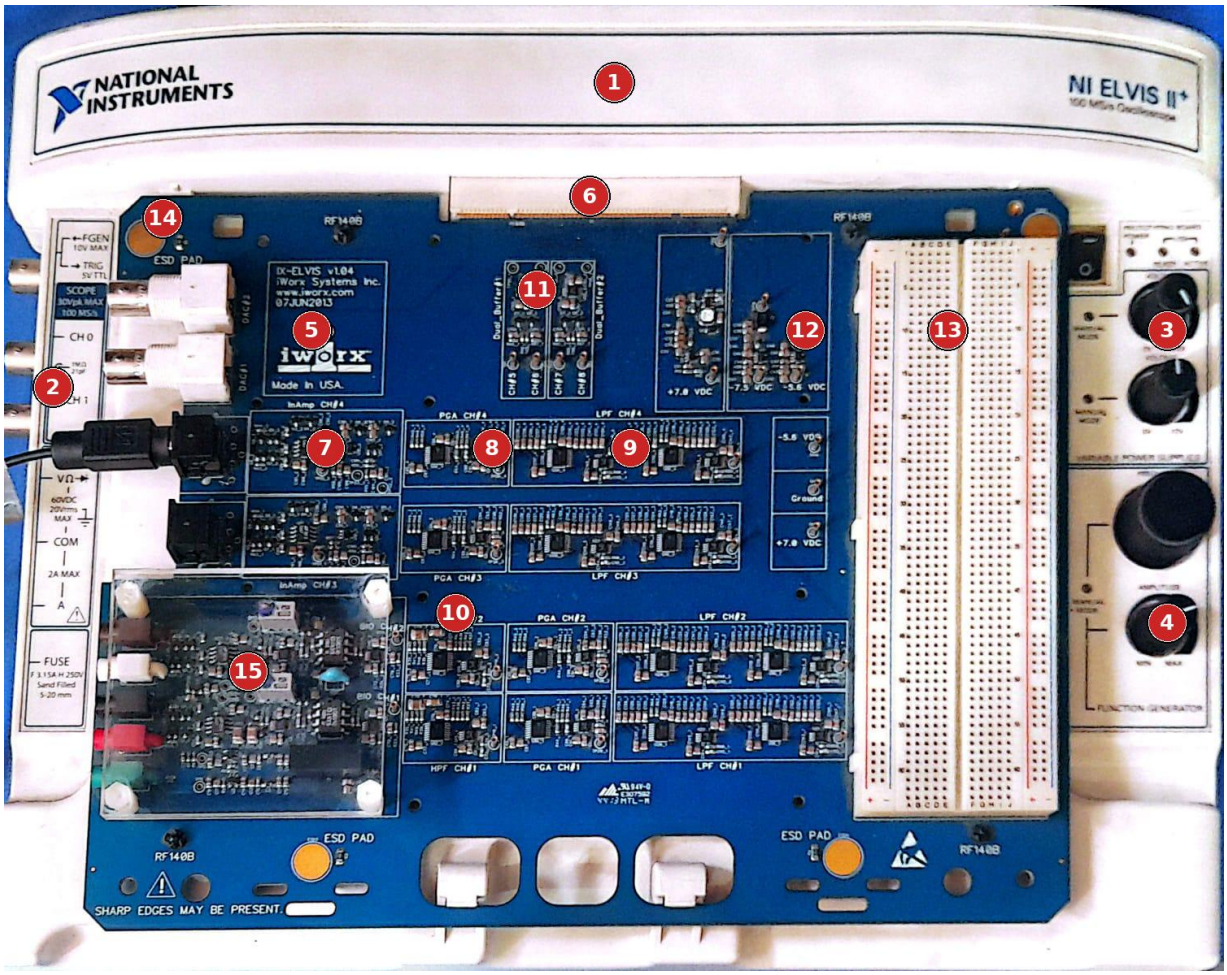


Figure 1.1: NI ELVIS II+ base unit with the IX-ELVIS interface board mounted on top, showing key functional components.

#	Component	Function
1	NI ELVIS II+ base unit	Multi-instrument platform (scope, DMM, power supplies, function generator) and USB link to PC
2	Prototyping board I/O panel	FGEN, TRIG, SCOPE (CH0/CH1), VΩ, COM, A, and FUSE connectors for the built-in instruments
3	Variable Power Supplies	Adjustable DC voltage supply knobs for powering auxiliary circuitry
4	Function Generator	Adjustable waveform generator controls (frequency/amplitude)
5	IX-ELVIS interface board	iWorx Systems analog front-end board mounted on the ELVIS prototyping area
6	Ribbon cable connector	Links the IX-ELVIS board electrically to the NI ELVIS II+ base unit
7	InAmp (Instrumentation Amplifier)	First-stage amplification of the raw sensor signal
8	PGA (Programmable Gain Amplifier)	Adjustable-gain second amplification stage
9	LPF (Low-Pass Filter)	Hardware-level removal of high-frequency noise per channel
10	HPF (High-Pass Filter)	Hardware-level removal of DC offset/baseline drift per channel
11	Dual Buffer 1/2	Signal buffering circuitry for Channels 5-8
12	Power regulation rails	Onboard +7.0 VDC / -7.5 VDC / -5.6 VDC / Ground supply for analog circuitry
13	Prototyping breadboard	Area for adding custom circuitry alongside fixed sensor-conditioning stages
14	ESD protection pad	Electrostatic discharge safety/grounding point
15	Sensor / BIO channel connector	Physical connection point for the iWorx sensor lead (ECG, PPG, spirometer, or dynamometer cable)

2. iWorx Sensors

iWorx manufactures physiological sensor modules designed for teaching and research in biomedical signal acquisition. Each sensor converts a specific physiological phenomenon into an analog voltage using a distinct sensing principle, which the IX-ELVIS board conditions and NI ELVIS II+ digitizes for MATLAB. The following sensors and channel assignments were used across the four experiments:

Experiment	Sensor	Signal Measured	NI ELVIS Channel
ECG	IX-ELVIS ECG electrodes	Heart's electrical activity	ai0 (Channel 1)
PPG	PTN-104	Blood volume pulse	ai2 (Channel 3)
Spirometer	SPN-304	Airflow / lung pressure	ai2 (Channel 3)
Hand Dynamometer	Force / strain-gauge sensor	Grip force	ai2

3. FT-220 Hand Dynamometer

The hand dynamometer sensor works on a ***resistive strain gauge*** principle, typically arranged in a Wheatstone bridge configuration. When the sensor's mechanical element is compressed by grip force, the strain gauge's electrical resistance changes proportionally to the applied force. The Wheatstone bridge circuit converts this tiny resistance change into a proportional output voltage, which is then amplified and conditioned before being digitized. Because this signal is mechanical/force-based rather than physiological/electrical, it changes far more slowly than ECG or PPG signals and requires a comparatively strong, low-noise voltage output.

Applications

- Stroke rehabilitation: tracking grip-strength recovery over time in stroke patients as an objective measure of motor recovery
- Physical therapy and rehabilitation progress tracking after hand/arm injury or surgery

- Sports and athletes strength assessment
- Screening for frailty and age-related muscle decline in elderly patients
- Occupational ergonomics assessing capacity for physically demanding tasks



Fig: Assembled FT-220 hand dynamometer, A-BT-220 tubing, and GPSN-100 pressure sensor

3.1 Methodology

- Sensor connected to channel ai2 on Dev1, sampled at 1000 Hz.
- Continuous background acquisition using startBackground with a DataAvailable listener, processing data in ~100 ms chunks.
- Real-time GUI built using uifigure/uiaxes, displaying live force waveform, current force, peak force, and windowed mean.
- Signal processing: 4th-order Butterworth low-pass filter (2 Hz cutoff) combined with a secondary 0.2 s moving-average smoother, appropriate since grip force is a slow-changing mechanical signal.
- Tare/zero function to correct baseline drift without requiring full recalibration.
- Two-point calibration wizard: captured a zero-force voltage (no grip) and a known-weight voltage (1 kg household reference item), then automatically computed the voltage-to-kilogram conversion factor.

3.2 Problems Faced and Troubleshooting

Problem 1: Aliasing-induced noise from low sampling rate

At an initial sampling rate of 100 Hz, the force signal exhibited a slow oscillatory "beating" pattern due to aliasing from 50 Hz mains interference occurring at the Nyquist limit. Increasing the sampling rate to 1000 Hz and applying a low-pass filter eliminated the artifact, resulting in a stable and accurate signal.

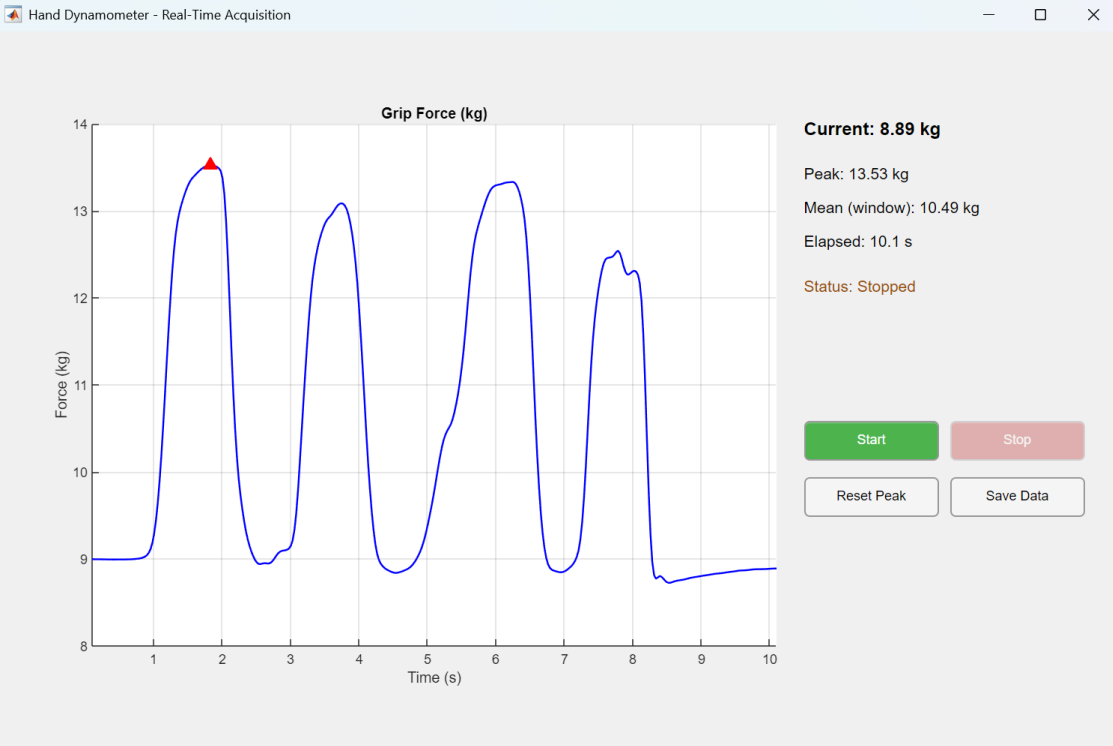
Problem 2: Visual noise exaggeration due to autoscaling

The resting baseline initially measured approximately 9 kg instead of 0 kg due to incorrect calibration. This was corrected using the tare function and a two-point calibration procedure.

3.3 Results — Before Calibration

Before calibration, the resting baseline measured approximately 9 kg instead of 0 kg due to a placeholder voltage-to-force conversion. As a result, the recorded peak force (13.53 kg) was not an accurate force value, although the grip pattern remained valid.

Subject Profile	
Weight	48 kg
Height	5'4"
BMI	18.2kg/m^2



3.4 Calibration Procedure

A two-point calibration was performed: (1) baseline voltage captured with no grip applied ("Cal: Set Zero"), (2) a 1 kg reference weight applied and its voltage captured ("Cal: Known Weight"). The app then computed CALIBRATION_FACTOR and CALIBRATION_OFFSET automatically from these two points.

3.5 Results — After Calibration

After the two-point calibration, the resting baseline correctly settled near 0 kg, and the peak grip force measured 17.21 kg, a genuine, calibrated force value. The calibration status confirmed a conversion factor of 18.406 kg/V derived from the 1 kg reference weight, giving confidence that the displayed numbers now reflect real applied force rather than an arbitrary placeholder scale.

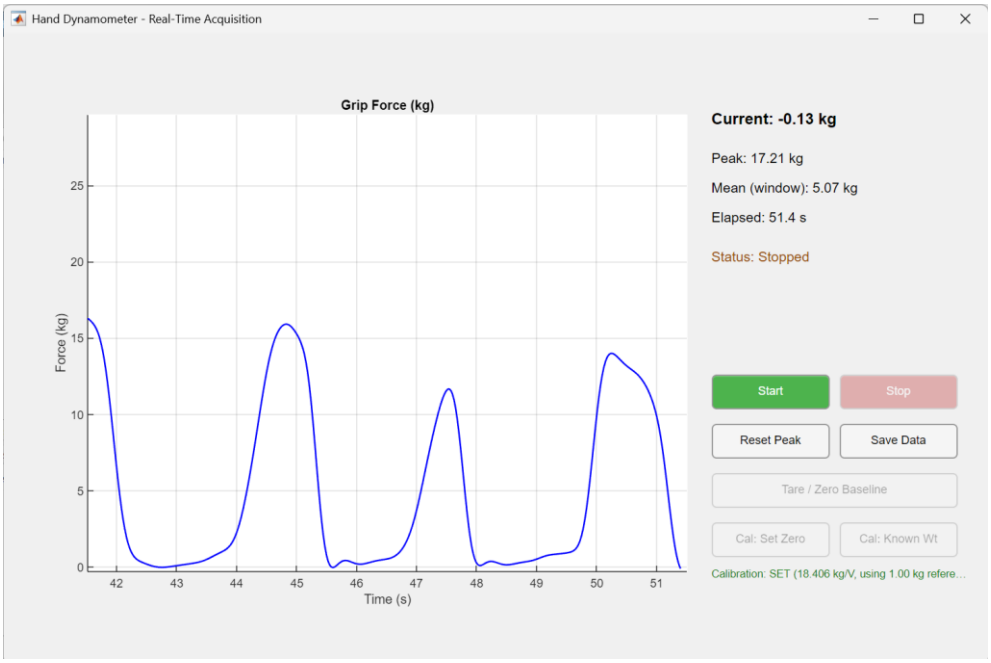


Figure Real-time hand dynamometer acquisition interface displaying grip force versus time during repeated voluntary grip contractions

4. PTN-104 Pulse plethysmography

The PTN-104 operates using pulse plethysmography (PPG), where an LED illuminates the fingertip and a photodetector measures changes in light caused by pulsatile blood volume. These changes are converted into an analog voltage signal that corresponds to the heartbeat and is recorded by MATLAB.

Applications

- Pulse oximetry combined with a second light wavelength, PPG is the basis of SpO2 (blood oxygen) monitoring
- Wearable fitness trackers and smartwatches for continuous heart-rate monitoring
- Remote/telehealth patient monitoring
- Stress and heart-rate-variability (HRV) assessment



Figure: The PTN-104 pulse plethysmograph

4.1 Methodology

- Sensor connected to channel ai2 on Dev1, sampled at 1000 Hz.
- Bandpass filter (0.5-8 Hz, 4th-order IIR) applied to isolate the slow pulsatile PPG waveform.
- Beats-per-minute (BPM) calculated from inter-beat intervals between successive detected pulse peaks (peak detection via findpeaks with prominence and minimum-distance constraints).
- Physiological state classified using BPM thresholds calibrated from the user's own recorded sessions: Resting State (75-87 BPM), Controlled Breathing (88-97 BPM), Active State (>=98 BPM).
- State classification averaged over a rolling window (stateWindowSec) and suppressed for the first 10 seconds of acquisition to avoid unstable early readings.

4.2 Problems Faced

Session-lock issue

The DAQ session would occasionally become "stuck," remaining locked from a previous acquisition attempt and preventing a new session from starting. This was resolved by implementing a releaseSession() helper function, combined with daqreset, which is called before every new acquisition attempt to guarantee a clean hardware session.

4.3 Results

Two physiological states were successfully captured and classified using the calibrated BPM thresholds:

Resting State

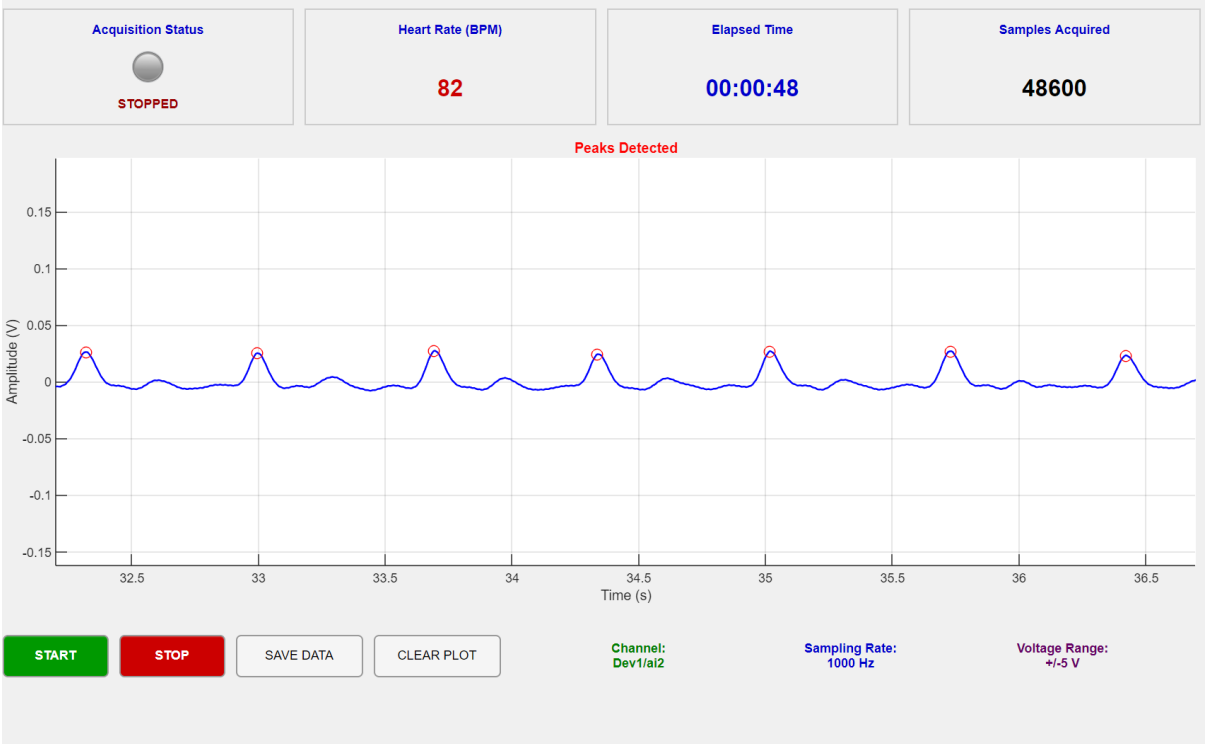


Figure 4.1: PPG waveform and detected pulse peaks — Resting State, Heart Rate: 82 BPM.

Manual BPM Verification (Rest State)

To verify the automated peak-detection algorithm, BPM was also calculated manually from five consecutive peak-to-peak intervals read directly off the rest-state waveform:

Interval #	Peak-to-Peak Time (s)
1	32.3
2	33.0
3	33.7
4	34.3
5	35.0
6	35.7
7	36.4

No. of beats = 6 intervals **Average IBI** = 36.4-32.3/6 = 0.683 sec/beat

Manual BPM (60 / avg IBI): 88 bpm

This approximately matches the software-calculated resting heart rate of 82 BPM , confirming the peak-detection algorithm is accurate.

Active State

Subject Profile	
Weight	48 kg
Height	5'4"
BMI	18.2kg/m^2
Jumping Jacks	40

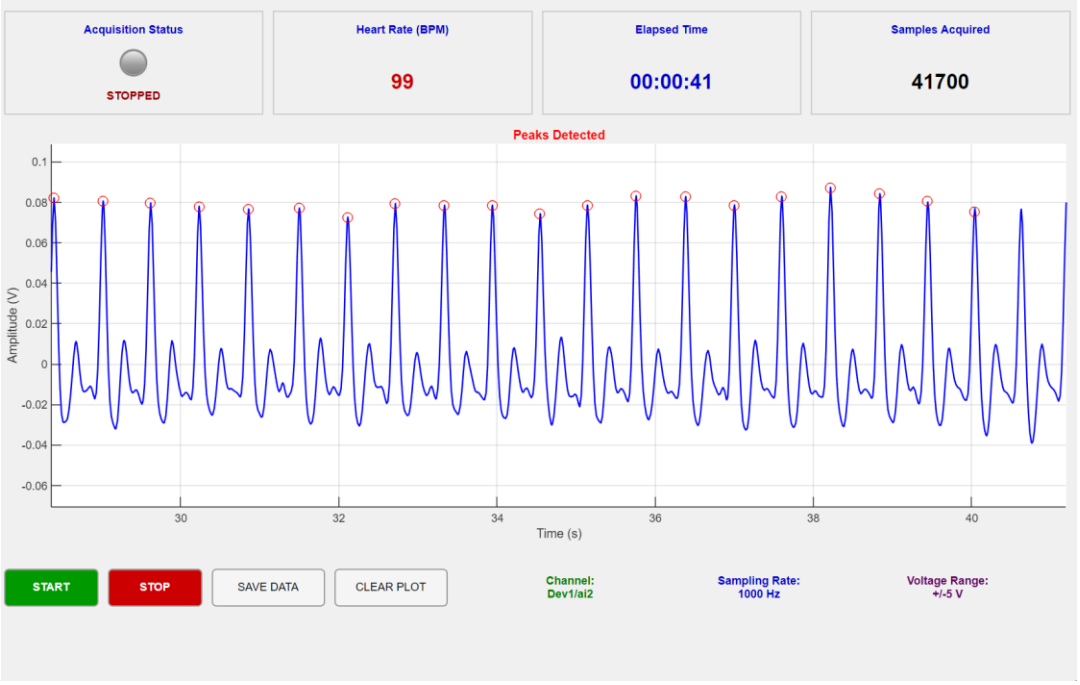


Figure 4.3: PPG waveform — Active State, Heart Rate: 99 BPM.

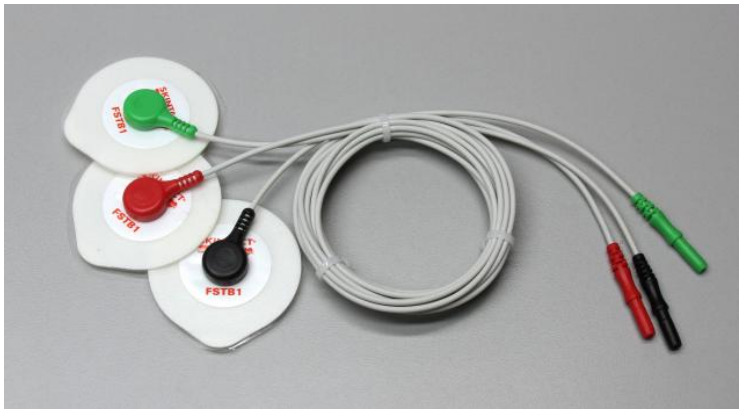
Condition	Heart Rate (BPM)	Classified State
Rest	82	RESTING STATE
Active	99	ACTIVE STATE

5. ECG (Electrocardiogram)

Every heartbeat begins with an electrical depolarization wave spreading through the heart muscle. This conducts through body tissue and appears as a tiny (millivolt-scale) voltage difference between skin electrodes at different body locations.

Applications

- Diagnosis of arrhythmias and other heart rhythm abnormalities
- Continuous cardiac monitoring during surgery or in the ICU
- Fitness wearables tracking heart rhythm during exercise
- Remote/telehealth cardiac screening



Electrode Placement

Lead Wire	Polarity	Electrode Placement
Red	+1 (Positive)	Left wrist
Black	−1 (Negative)	Right wrist
Green	C (Ground)	Right ankle

Figure Disposable three-lead ECG electrode assembly used for biopotential signal acquisition during ECG measurements.

5.1 Methodology

- Sensor: IX-ELVIS ECG electrodes -> analog input ai0 (Dev1), sampled at 1000 Hz
- Electrode placement: Red - left wrist | Black - right wrist | Green - right ankle (ground reference)
- Skin preparation: skin cleaned with an alcohol swab, then dried with gauze before electrode placement, to reduce skin-electrode impedance and improve signal quality
- Filtering: Butterworth bandpass (0.5-40 Hz) + 50 Hz IIR notch (mains) + ~10 ms moving-average smoothing + z-score normalization
- R-peak detection via findpeaks; classification from RR-interval statistics (Normal Sinus Rhythm / Bradycardia / Tachycardia / Irregular)

5.2 Problems Faced

Persistent 50 Hz mains noise

The raw ECG signal showed persistent high-frequency contamination. FFT analysis performed on saved .mat files confirmed a distinct spike at 50 Hz, identifying mains electrical interference as the specific noise source rather than a random hardware fault.

Suspected missing earth ground

Investigation pointed toward a missing or improper earth ground connection on the acquisition setup as the likely root cause of the persistent noise a hardware-level issue that cannot be fully resolved through software filtering alone. The human body acts as an antenna for ambient 50 Hz interference when electrodes are attached to skin, a physical effect that cannot be corrected purely in software.

AC unit interference

The room's air-conditioning unit was identified as an additional noise source through capacitive coupling, compounding the mains interference problem.

Iterative acquisition redesign

Verify earth grounding using a "touch test", comparing the 50 Hz FFT peak before and after touching a grounded metal object while recording. The acquisition method evolved from an initial fixed-duration recording approach to continuous background acquisition by ensuring a proper ground connection at the electrode/board level, combined with the bandpass + notch filtering pipeline described above together these produced a clean, classifiable waveform.

FFT Diagnosis

To pin down the exact cause of the persistent noise, a Fast Fourier Transform (FFT) was performed on the raw saved ECG data. FFT converts a signal from the time domain (voltage vs. time, which is what the oscilloscope-style plot shows) into the frequency domain (magnitude vs. frequency), revealing exactly which frequencies are present and how strong each one is information that is impossible to read directly off the noisy time-domain waveform.

Zooming the FFT plot into the 45-55 Hz range (shown below) revealed a sharp, dominant magnitude spike centered almost exactly at 50 Hz, far larger than any neighboring frequency component. This is the textbook frequency-domain signature of mains electrical interference. Its presence confirmed definitively that the contamination was electrical in origin (i.e. coming from the room's AC power supply/wiring) rather than muscle artifact, subject movement, or a faulty sensor - ruling out several other possible causes in one step.

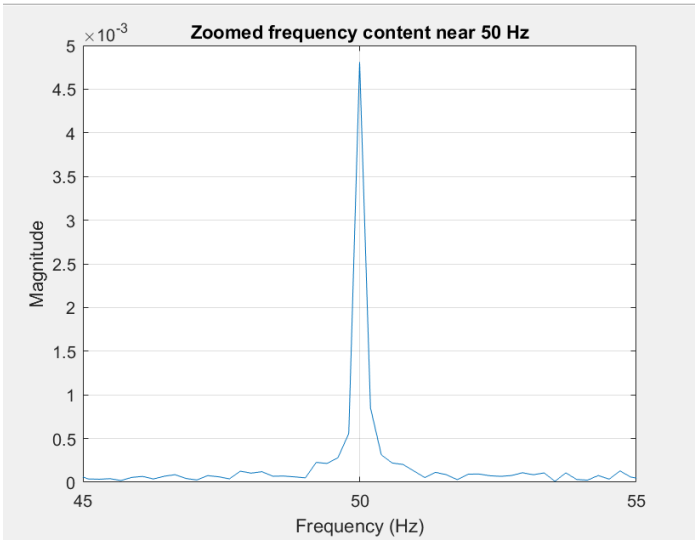


Figure Zoomed FFT magnitude spectrum of the raw ECG signal near 50 Hz, showing the sharp interference peak at the mains frequency.

Touch-Pad Grounding Test

With mains interference confirmed as the noise source via FFT, the next question was why it was so strong i.e. whether the acquisition setup had an adequate earth ground. To test this directly, a simple touch-pad test was performed: while continuously recording, the subject touched the ESD grounding pad on the IX-ELVIS board (a verified, true earth-ground point on the hardware) with their free hand, while keeping the ECG electrodes attached as normal.

The 50 Hz interference visibly and immediately dropped in amplitude for as long as contact with the grounding pad was maintained, and returned once contact was released. This is a strong practical confirmation that, without this extra ground contact, the subject's body was acting as an unintentional antenna, picking up ambient 50 Hz fields radiating from room wiring and the nearby AC unit because the acquisition setup's own ground reference was too weak on its own to fully sink this interference.

Solution Implemented

Based on these two diagnostic steps (FFT + touch-pad test), the fix applied was two-fold: (1) a stable, low-impedance earth-ground connection was established between the subject and the instrumentation ground (2) this was combined with the existing software bandpass (0.5-40 Hz) and 50 Hz notch filtering pipeline to clean up any small residual interference left over. Only after combining a hardware-level fix (proper grounding) with a software-level fix (filtering) was a consistently clean, reliably classifiable ECG waveform achieved neither approach alone was fully sufficient.

5.3 Results

Despite the mains noise present in the raw signal, the filtering pipeline successfully recovered a clean, classifiable ECG waveform with consistent R-peak detection.

Subject Profile	
Height	5'4"
Weight	50 kg
BMI	18.9 kg/m^2

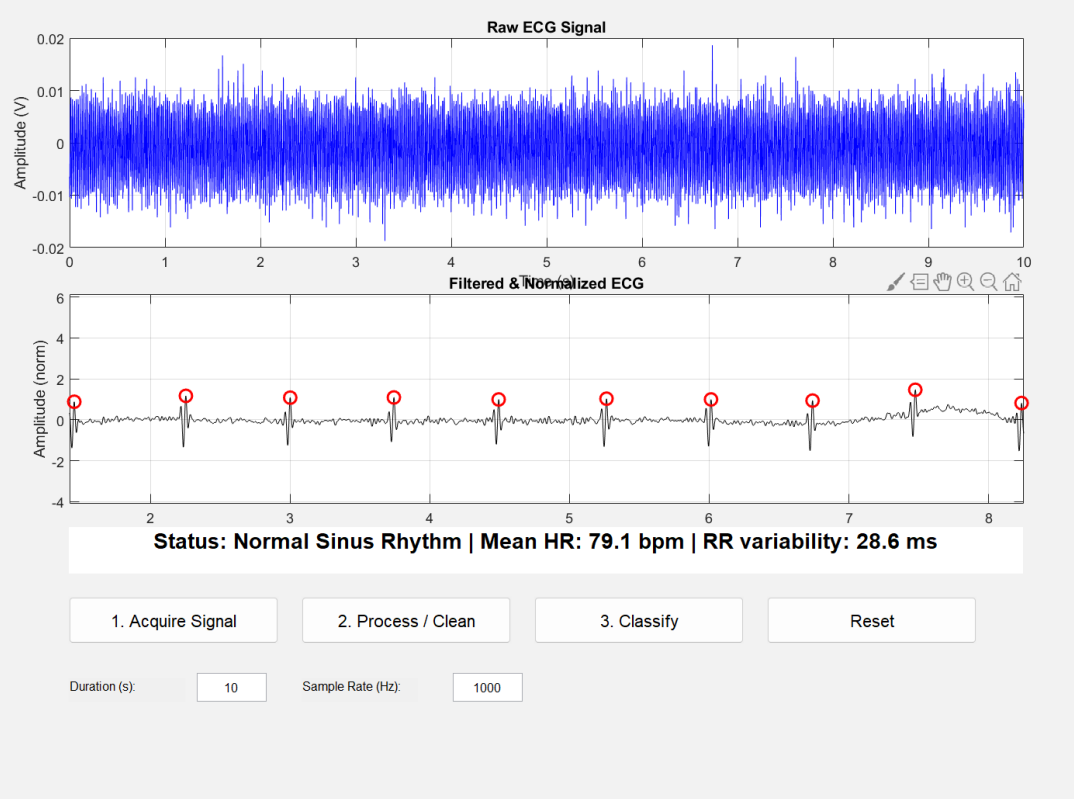


Figure 5.1: Raw ECG signal (top) and filtered/normalized ECG with detected R-peaks (bottom).

After applying the bandpass, notch, and smoothing filters, the noisy raw ECG signal (top trace) was successfully cleaned into a recognizable, regularly beating waveform (bottom trace) with clearly detected R-peaks. The system classified the rhythm as Normal Sinus Rhythm, with a mean heart rate of 79.1 bpm and an RR-interval variability of 28.6 ms, both consistent with a healthy, regular heartbeat and confirming that the filtering pipeline successfully recovered a clean, classifiable signal despite the significant mains interference visible in the raw trace.

6. SPN-304 Spirometer Experiment

The SPN-304 measures the pressure difference created as air flows across a resistive element in the flowhead. This pressure is proportional to airflow rate; integrating airflow over time yields lung volume.

Applications

- Diagnosis and monitoring of asthma, COPD, and other respiratory conditions
- Pre-surgical pulmonary function assessment
- Athletic performance and lung-capacity evaluation
- Occupational health screening (e.g. exposure to dust/chemicals)

6.1 Methodology

- Sensor: SPN-304, connected via IX-ELVIS Channel 3 -> NI ELVIS II+ analog input ai2 (Dev1)
- Sampling rate: 500 Hz, 60-second fixed-duration recording per condition
- Bandpass filter (0.05-5 Hz, 3rd-order Butterworth) applied to isolate breathing-relevant content and remove drift plus high-frequency noise.
- Voltage converted to pressure (mmH2O) using the sensor's manufacturer-specified calibration constant (25.7 mmH2O per volt).
- Airflow reported in pressure-proportional arbitrary units (a.u.), since true L/s conversion requires a known-volume calibration syringe which was not available.
- Volume obtained via cumulative trapezoidal integration (cumtrapz) of airflow, with linear detrending to remove slow baseline drift common in spirometry volume traces.
- Breath detection via peak/trough detection (findpeaks) on the volume signal, with prominence auto-scaled to the observed signal amplitude.
- Per-breath metrics computed: tidal volume (TV), respiratory rate (RR), inspiration/expiration time (Ti/Te), I:E ratio, and peak inspiratory/expiratory flow (PIF/PEF).

6.2 Problems Faced

Trough-peak pairing bug

The initial logic used to pair each breath's trough-peak-trough cycle for tidal volume calculation was flawed, leading to incorrect per-breath metrics. This was identified and corrected to properly segment individual breath cycles.

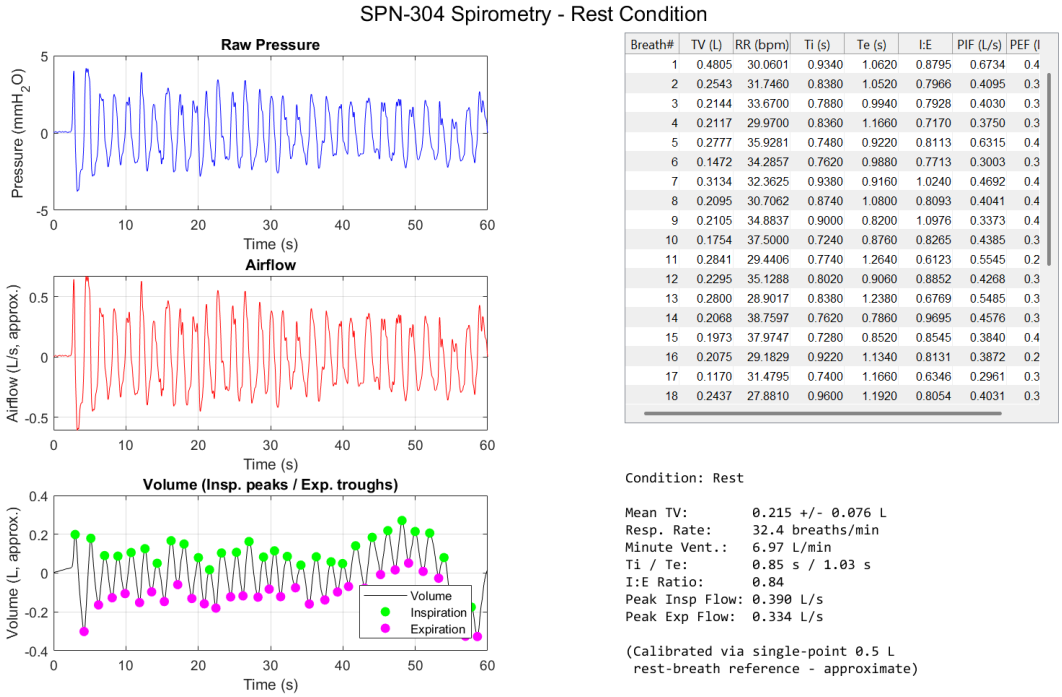
Calibration limitation

Volume and airflow values are reported in arbitrary units (a.u.) rather than absolute liters, since no calibration syringe of known volume was available. Volume and airflow measurements were calibrated using a standard 1 L volume to convert the sensor output into liters (L) and liters per second (L/s). This calibration provides meaningful quantitative measurements for comparing different breathing conditions (rest, deep breathing, and post-exercise). However, as the calibration was performed using a simple reference rather than a clinical calibration syringe, the results are intended for experimental analysis and should not be interpreted as clinical-grade measurements.

Note on standard industry practice: clinical and industrial spirometers are normally calibrated using a dedicated 3-litre calibration syringe, which injects a precisely known volume of air into the flowhead so the device's voltage-to-volume conversion can be verified against ground truth. This 3 L syringe calibration is the accepted standard method in clinical practice. A 3 L calibration syringe was not available for this setup, so the 1 L reference method described above was used instead as a practical substitute; a proper 3 L syringe calibration remains the recommended upgrade for future, clinically-accurate volume readings.

6.3 Results

Condition 1: Rest



Breath#	TV (L)	RR (bpm)	Ti (s)	Te (s)	I:E	PIF (L/s)	PEF (l
1	0.4805	30.0601	0.9340	1.0620	0.8795	0.6734	0.4
2	0.2543	31.7460	0.8380	1.0520	0.7966	0.4095	0.3
3	0.2144	33.6700	0.7880	0.9940	0.7928	0.4030	0.3
4	0.2117	29.9700	0.8360	1.1660	0.7170	0.3750	0.3
5	0.2777	35.9281	0.7480	0.9220	0.8113	0.6315	0.4
6	0.1472	34.2857	0.7620	0.9880	0.7713	0.3003	0.3
7	0.3134	32.3625	0.9380	0.9160	1.0240	0.4692	0.4
8	0.2095	30.7062	0.8740	1.0800	0.8093	0.4041	0.4
9	0.2105	34.8837	0.9000	0.8200	1.0976	0.3373	0.4
10	0.1754	37.5000	0.7240	0.8760	0.8265	0.4385	0.3
11	0.2841	29.4406	0.7740	1.2640	0.6123	0.5545	0.2
12	0.2295	35.1288	0.8020	0.9060	0.8852	0.4268	0.3
13	0.2800	28.9017	0.8380	1.2380	0.6769	0.5485	0.3
14	0.2068	38.7597	0.7620	0.7860	0.9695	0.4576	0.3
15	0.1973	37.9747	0.7280	0.8520	0.8545	0.3840	0.4
16	0.2075	29.1829	0.9220	1.1340	0.8131	0.3872	0.2
17	0.1170	31.4795	0.7400	1.1660	0.6346	0.2961	0.3
18	0.2437	27.8810	0.9600	1.1920	0.8054	0.4031	0.3

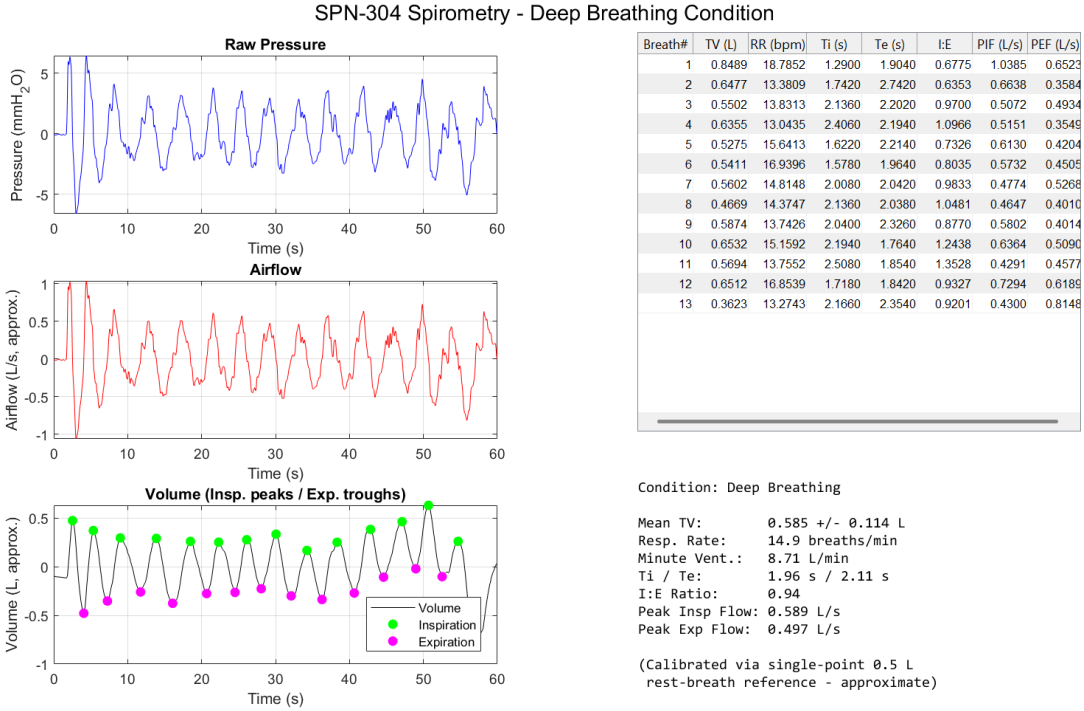
```
Condition: Rest

Mean TV:         0.215 +/- 0.076 L
Resp. Rate:      32.4 breaths/min
Minute Vent.:    6.97 L/min
Ti / Te:         0.85 s / 1.03 s
I:E Ratio:       0.84
Peak Insp Flow:  0.390 L/s
Peak Exp Flow:   0.334 L/s

(Calibrated via single-point 0.5 L
 rest-breath reference - approximate)
```

Figure 6.1: Rest condition — raw pressure, airflow, and volume traces with per-breath metrics table.

Condition 2: Deep Breathing



Breath#	TV (L)	RR (bpm)	Ti (s)	Te (s)	I:E	PIF (L/s)	PEF (L/s)
1	0.8489	18.7852	1.2900	1.9040	0.6775	1.0385	0.6523
2	0.6477	13.3809	1.7420	2.7420	0.6353	0.6638	0.3584
3	0.5502	13.8313	2.1360	2.2020	0.9700	0.5072	0.4934
4	0.6355	13.0435	2.4060	2.1940	1.0966	0.5151	0.3549
5	0.5275	15.6413	1.6220	2.2140	0.7326	0.6130	0.4204
6	0.5411	16.9396	1.5780	1.9640	0.8035	0.5732	0.4505
7	0.5602	14.8148	2.0080	2.0420	0.9833	0.4774	0.5268
8	0.4669	14.3747	2.1360	2.0380	1.0481	0.4647	0.4010
9	0.5874	13.7426	2.0400	2.3260	0.8770	0.5802	0.4014
10	0.6532	15.1592	2.1940	1.7640	1.2438	0.6364	0.5090
11	0.5694	13.7552	2.5080	1.8540	1.3528	0.4291	0.4577
12	0.6512	16.8539	1.7180	1.8420	0.9327	0.7294	0.6189
13	0.3623	13.2743	2.1660	2.3540	0.9201	0.4300	0.8148

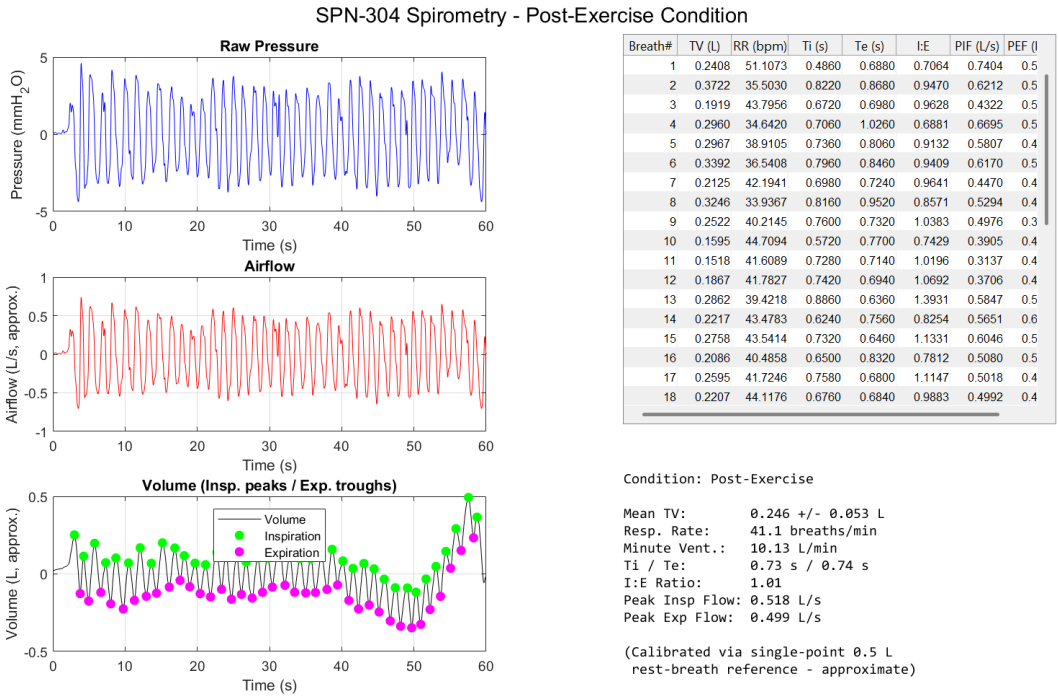
```
Condition: Deep Breathing

Mean TV:         0.585 +/- 0.114 L
Resp. Rate:      14.9 breaths/min
Minute Vent.:    8.71 L/min
Ti / Te:         1.96 s / 2.11 s
I:E Ratio:       0.94
Peak Insp Flow:  0.589 L/s
Peak Exp Flow:   0.497 L/s

(Calibrated via single-point 0.5 L
 rest-breath reference - approximate)
```

Figure 6.2: Deep breathing condition — raw pressure, airflow, and volume traces with per-breath metrics table.

Condition 3: Post-Exercise



Breath#	TV (L)	RR (bpm)	Ti (s)	Te (s)	I:E	PIF (L/s)	PEF (l
1	0.2408	51.1073	0.4860	0.6880	0.7064	0.7404	0.5
2	0.3722	35.5030	0.8220	0.8680	0.9470	0.6212	0.5
3	0.1919	43.7956	0.6720	0.6980	0.9628	0.4322	0.5
4	0.2960	34.6420	0.7060	1.0260	0.6881	0.6695	0.5
5	0.2967	38.9105	0.7360	0.8060	0.9132	0.5807	0.4
6	0.3392	36.5408	0.7960	0.8460	0.9409	0.6170	0.5
7	0.2125	42.1941	0.6980	0.7240	0.9641	0.4470	0.4
8	0.3246	33.9367	0.8160	0.9520	0.8571	0.5294	0.4
9	0.2522	40.2145	0.7600	0.7320	1.0383	0.4976	0.3
10	0.1595	44.7094	0.5720	0.7700	0.7429	0.3905	0.4
11	0.1518	41.6089	0.7280	0.7140	1.0196	0.3137	0.4
12	0.1867	41.7827	0.7420	0.6940	1.0692	0.3706	0.4
13	0.2862	39.4218	0.8860	0.6360	1.3931	0.5847	0.5
14	0.2217	43.4783	0.6240	0.7560	0.8254	0.5651	0.6
15	0.2758	43.5414	0.7320	0.6460	1.1331	0.6046	0.5
16	0.2086	40.4858	0.6500	0.8320	0.7812	0.5080	0.5
17	0.2595	41.7246	0.7580	0.6800	1.1147	0.5018	0.4
18	0.2207	44.1176	0.6760	0.6840	0.9883	0.4992	0.4

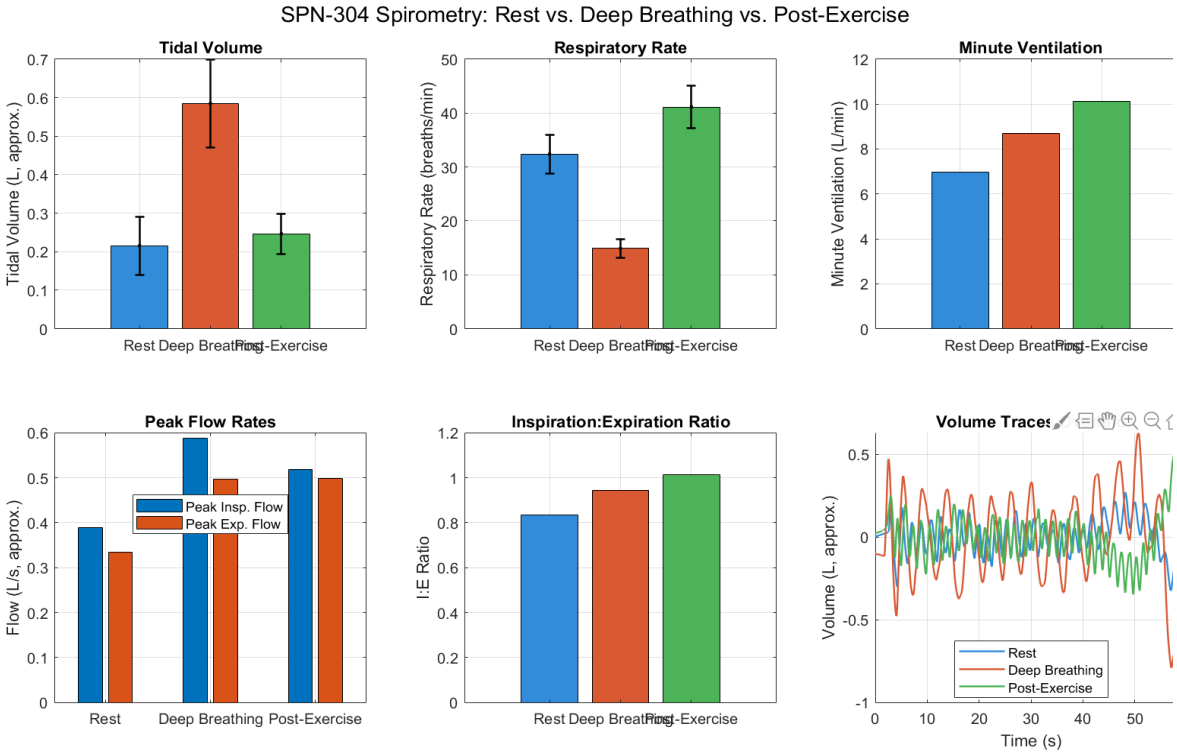
```
Condition: Post-Exercise

Mean TV:         0.246 +/- 0.053 L
Resp. Rate:      41.1 breaths/min
Minute Vent.:    10.13 L/min
Ti / Te:         0.73 s / 0.74 s
I:E Ratio:       1.01
Peak Insp Flow: 0.518 L/s
Peak Exp Flow:  0.499 L/s

(Calibrated via single-point 0.5 L
 rest-breath reference - approximate)
```

Figure 6.3: Post-exercise condition — raw pressure, airflow, and volume traces with per-breath metrics table.

Comparison across Conditions



Condition	Mean TV (L)	Resp. Rate (bpm)	Minute Vent. (L/min)	I:E Ratio
Rest	0.215	32.4	6.97	0.84
Deep Breathing	0.585	14.9	8.71	0.94
Post-Exercise	0.246	41.1	10.13	1.01

As expected physiologically, deep breathing produced the largest tidal volume with the lowest respiratory rate, while post-exercise breathing showed the highest respiratory rate and minute ventilation, consistent with the body's increased oxygen demand following physical exertion.

Common Noise Challenges

A recurring theme across experiments was 50 Hz mains electrical interference, which manifested differently depending on each signal's bandwidth and amplitude. It was straightforward to remove from the hand dynamometer signal (a slow mechanical signal well separated from 50 Hz) but persistently problematic in the ECG signal, where the noise frequency overlaps with genuine physiological content and the body itself acts as an antenna for ambient electrical fields.

Code Repository & Software Setup

All MATLAB code for the four experiments is available in the GitHub repository, organized by experiment with a README containing setup, usage, required MATLAB library installation, and hardware verification instructions are also given in repository.

https://github.com/mariyamamin-hub/Biomedical-signal-acquisition-using-NI-ELVIS-i-Worx/tree/main/github_repo/github_repo